

Hash Table Collision Resolution Strategies

CSE207: Data Structure
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Problem Statement

Given the following keys:

224562, 137456, 214562, 140145, 214576, 162145, 144467, 199645, 234534

Insert them into a hash table of size 19 using the modulo-division hashing method $h(k) = k \bmod 19$, and apply the following collision resolution strategies:

1. Linear Probing: $h(k, i) = (h(k) + i) \bmod 19$
2. Quadratic Probing: $h(k, i) = (h(k) + i^2) \bmod 19$
3. Pseudorandom Probing: $h(k, i) = (h(k) + (3 \times i - 1)) \bmod 19$
4. Key-Offset Method: $h(k, i) = (h(k) + i \times \text{offset}) \bmod 19$, where $\text{offset} = k \bmod 7$

For each method, calculate the number of collisions. (*Reference: Project 19, Page 640, Data Structures: A Pseudocode Approach with C*)

Solution

Initial Hash Values

First, we calculate the initial hash position for each key using $h(k) = k \bmod 19$:

Method 1: Linear Probing

Rehash function: $h(k, i) = (h(k) + i) \bmod 19$

Method 2: Quadratic Probing

Rehash function: $h(k, i) = (h(k) + i^2) \bmod 19$

Method 3: Pseudorandom Probing

Rehash function: $h(k, i) = (h(k) + (3 \times i - 1)) \bmod 19$

Table 1: Initial Hash Positions

Key	$k \bmod 19$	Initial Position
224562	1	1
137456	10	10
214562	14	14
140145	1	1 (collision)
214576	9	9
162145	18	18
144467	10	10 (collision)
199645	12	12
234534	17	17

Table 2: Linear Probing Insertion Process

Key	Initial Hash	Probes	Collisions	Final Index
224562	1	1	0	1
137456	10	10	0	10
214562	14	14	0	14
140145	1	1, 2	1	2
214576	9	9	0	9
162145	18	18	0	18
144467	10	10, 11	1	11
199645	12	12	0	12
234534	17	17	0	17
Total Collisions:				2

Method 4: Key-Offset Method

Offset: $\text{offset} = k \bmod 7$

Rehash function: $h(k, i) = (h(k) + i \times \text{offset}) \bmod 19$

Table 3: Quadratic Probing Insertion Process

Key	Initial Hash	Probes	Collisions	Final Index
224562	1	1	0	1
137456	10	10	0	10
214562	14	14	0	14
140145	1	1, 2	1	2
214576	9	9	0	9
162145	18	18	0	18
144467	10	10, 11	1	11
199645	12	12	0	12
234534	17	17	0	17
Total Collisions:				2

Table 4: Pseudorandom Probing Insertion Process

Key	Initial Hash	Probes	Collisions	Final Index
224562	1	1	0	1
137456	10	10	0	10
214562	14	14	0	14
140145	1	1, 3	1	3
214576	9	9	0	9
162145	18	18	0	18
144467	10	10, 12	1	12
199645	12	12, 14, 17	2	17
234534	17	17, 0	1	0
Total Collisions:				5

Table 5: Key-Offset Method Insertion Process

Key	Offset	Initial Hash	Probes	Collisions	Final Index
224562	2	1	1	0	1
137456	4	10	10	0	10
214562	5	14	14	0	14
140145	5	1	1, 6	1	6
214576	5	9	9	0	9
162145	4	18	18	0	18
144467	1	10	10, 11	1	11
199645	5	12	12	0	12
234534	6	17	17	0	17
Total Collisions:				2	